

Practice problems #3

1. True or False. The following matrix has a zero eigenvalue.

$$A = \begin{bmatrix} 1 & -4 & 9 \\ -1 & 2 & -4 \\ 5 & -6 & 10 \end{bmatrix}$$

2. In the following, show that

- (a) the given λ is an eigenvalue of the given matrix A .
- (b) Find a basis for the eigenspace corresponding to λ .
- (c) What is the geometric multiplicity of λ ?
- (d) Give an eigenvector of A corresponding to λ other than your basis vectors.

(2-1)

$$A = \begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix}, \quad \lambda = 10$$

(2-2)

$$A = \begin{bmatrix} 4 & 2 & 3 \\ -1 & 1 & -3 \\ 2 & 4 & 9 \end{bmatrix}, \quad \lambda = 3$$

Answers or Hints on page 2.

#1: True. Why?

#2(a): Do something with $(A - \lambda I)$.

#2(b): There can be many different answers. As an example,

$$(2-1): \left\{ \begin{bmatrix} -1 \\ 3 \end{bmatrix} \right\}, \quad (2-2): \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

#2(c): (2-1) 1, (2-2) 2.

#2(d): There can be infinitely many different answers. As an example,

$$(2-1): \begin{bmatrix} -2 \\ 6 \end{bmatrix}, \quad (2-2): \begin{bmatrix} -5 \\ 1 \\ 1 \end{bmatrix}.$$